Project Title

Comprehensive Analysis and Dietary strategies with Tableau : A College Food Choices Case Study

Team Id : SWUID2026226489
SWUID2026226490
SWUID2026226275

Team Size: 3

Team Member : Ram Sai Vanapalli

Team Member : Naga Pushpa Latha Ravula

Team Member : Sudheeksha Sangana

1. INTRODUCTION

1.1 Project Overview

"The **Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study** is an interactive, data-driven web application designed to analyze the dietary habits and health awareness of college students. In the academic environment, food choices significantly impact student well-being, yet students often lack accessible information regarding nutritional values and consumption trends. Traditional methods of food tracking are manual, time-consuming, and fail to provide actionable insights. This project simplifies and automates the analysis process by leveraging Tableau's visualization capabilities integrated with a Flask-based web interface. The system collects and processes student dietary data, allowing users to explore food consumption patterns, health correlations, and nutritional trends through dynamic, interactive dashboards. By combining data analytics with web technology, this project demonstrates how visual intelligence can be applied to foster better nutritional awareness and promote healthier lifestyle choices among students.".

1.2 Purpose

"The primary purpose of the **Comprehensive Analysis and Dietary Strategies with Tableau** project is to bridge the gap between complex dietary data and student health awareness by automating the visualization process. Instead of relying on manual surveys or static reports, this system empowers students and administrators to analyze food intake patterns, lifestyle choices, and health outcomes in real-time. This approach minimizes human error, reduces data processing time, and provides clear, data-driven insights that are easy to interpret.

Another core objective is to demonstrate the practical application of Business Intelligence (BI) tools in the education and wellness sector. The system provides an interactive platform where students can visualize their eating habits and understand the correlation between food choices and daily physical activity. Additionally, this project enhances technical proficiency in data cleaning, Flask web framework integration, and Tableau dashboard design. The platform is designed to be scalable, offering a foundation for future enhancements such as AI-driven personalized diet recommendations and integration with broader institutional wellness programs, making it a sustainable solution for fostering a healthier campus environment."

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Ideation Phase
Brainstorm & Idea Prioritization Template

Date	30 Jun 2026
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Maximum Marks	4 Marks

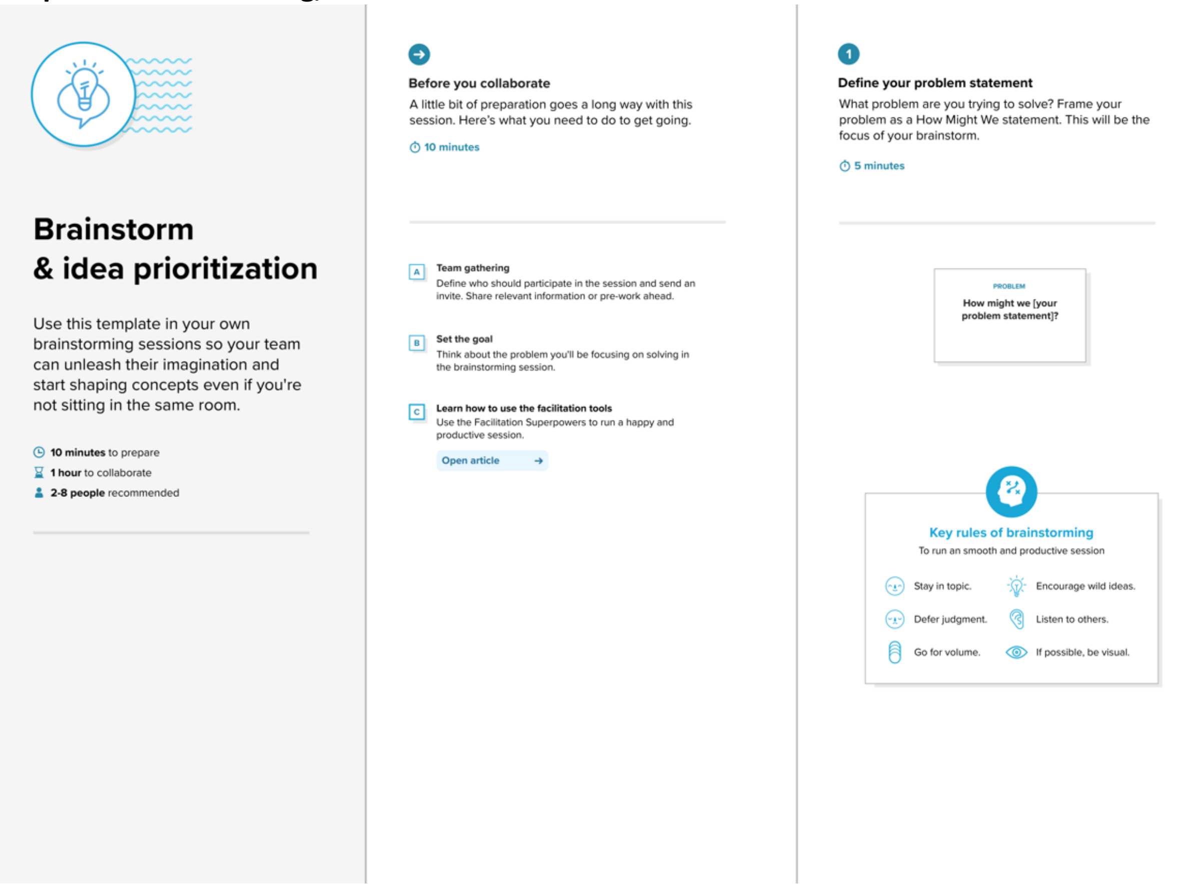
Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

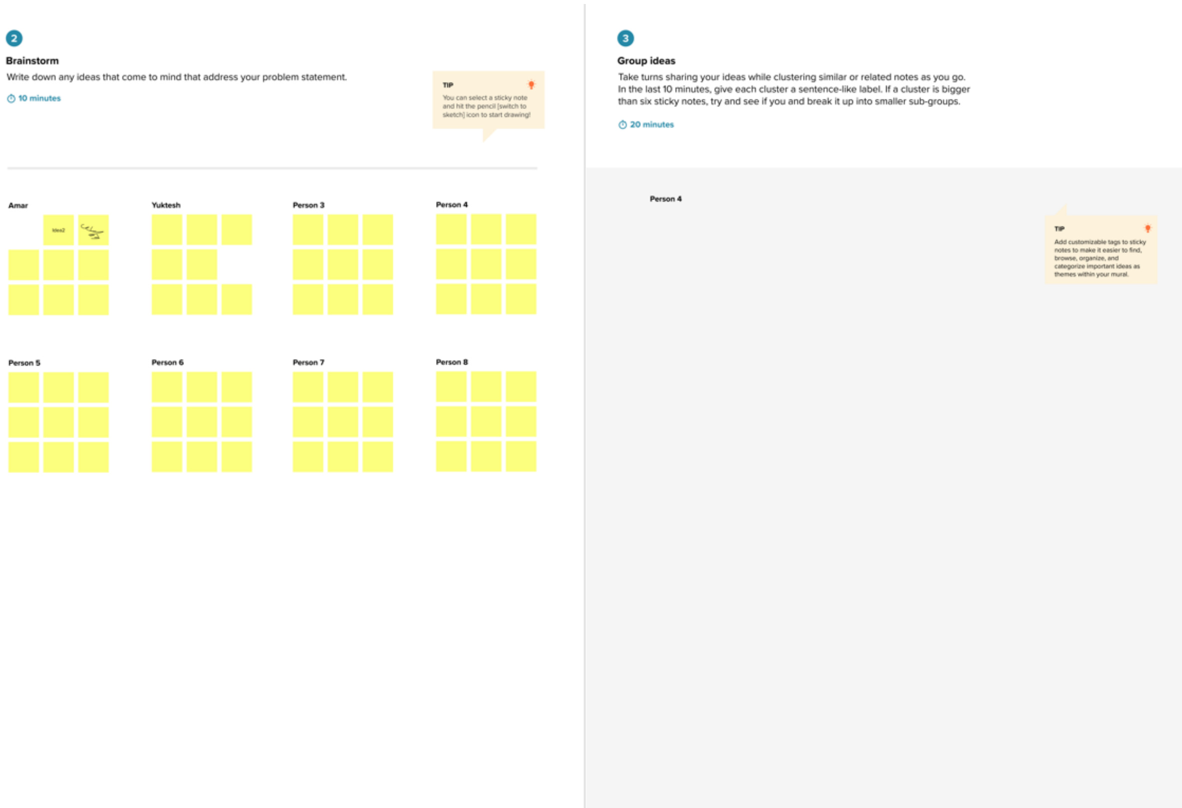
Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping

Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization

Ideation Phase
Define the Problem Statements

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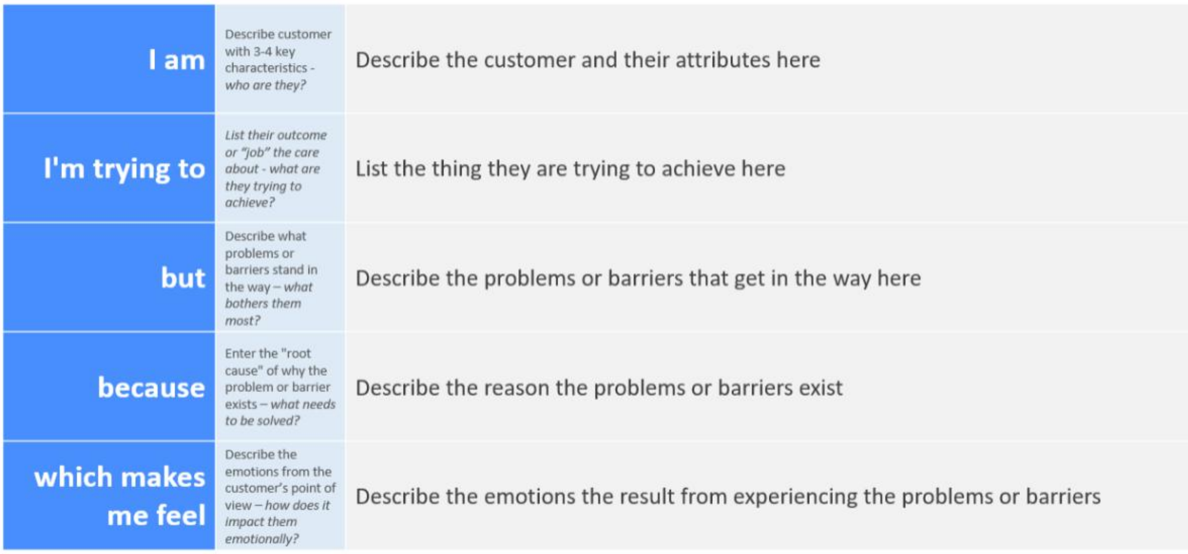
Customer Problem Statement Template:
Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.
A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.



I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

Reference: https://miro.com/templates/customer-problem-statement/

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A college student	Choose healthy food options in canteen	There is a lack of nutritional information	The current menu doesn't show calorie or health data	Confused about my diet
PS-2	Track my daily food intake habits	Track my daily food intake habits	It's hard to visualize my eating patterns	Manual tracking is time consuming and difficult	Frustrated with my lifestyle

Ideation Phase
Empathize & Discover

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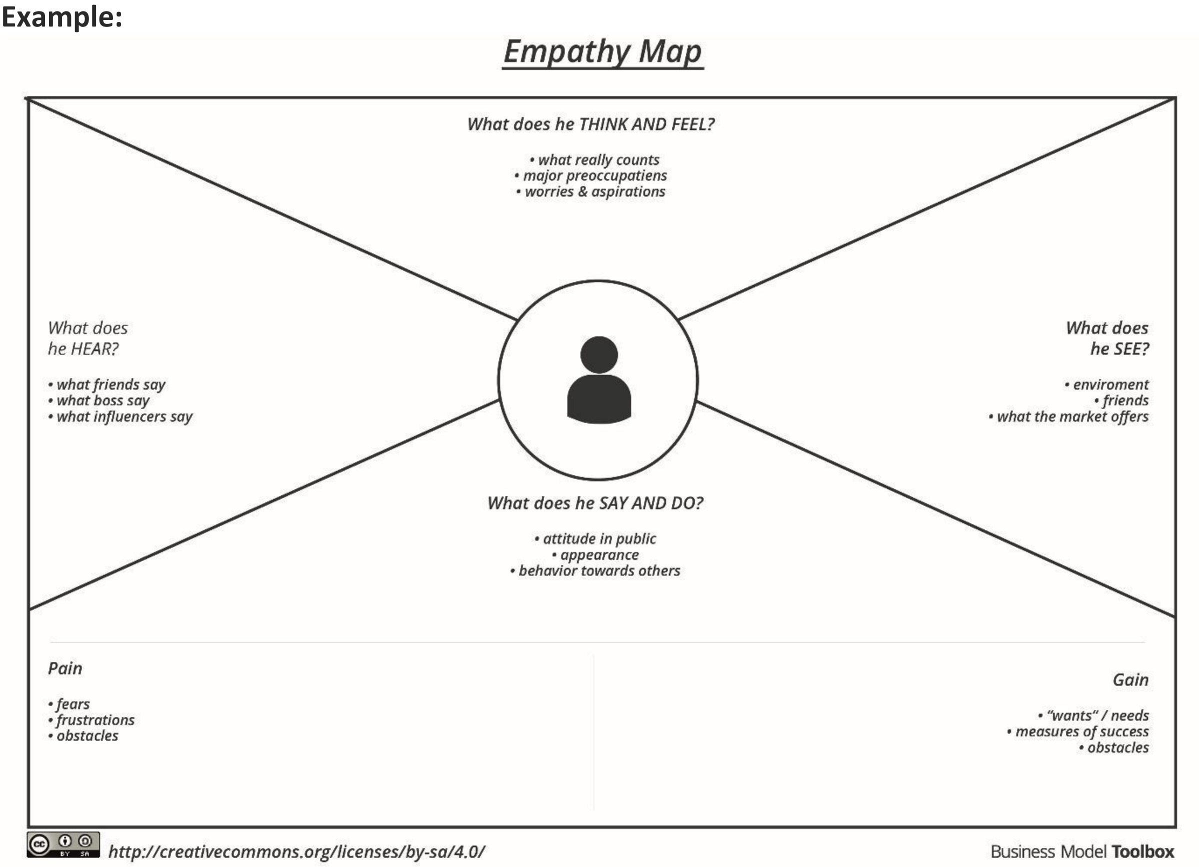
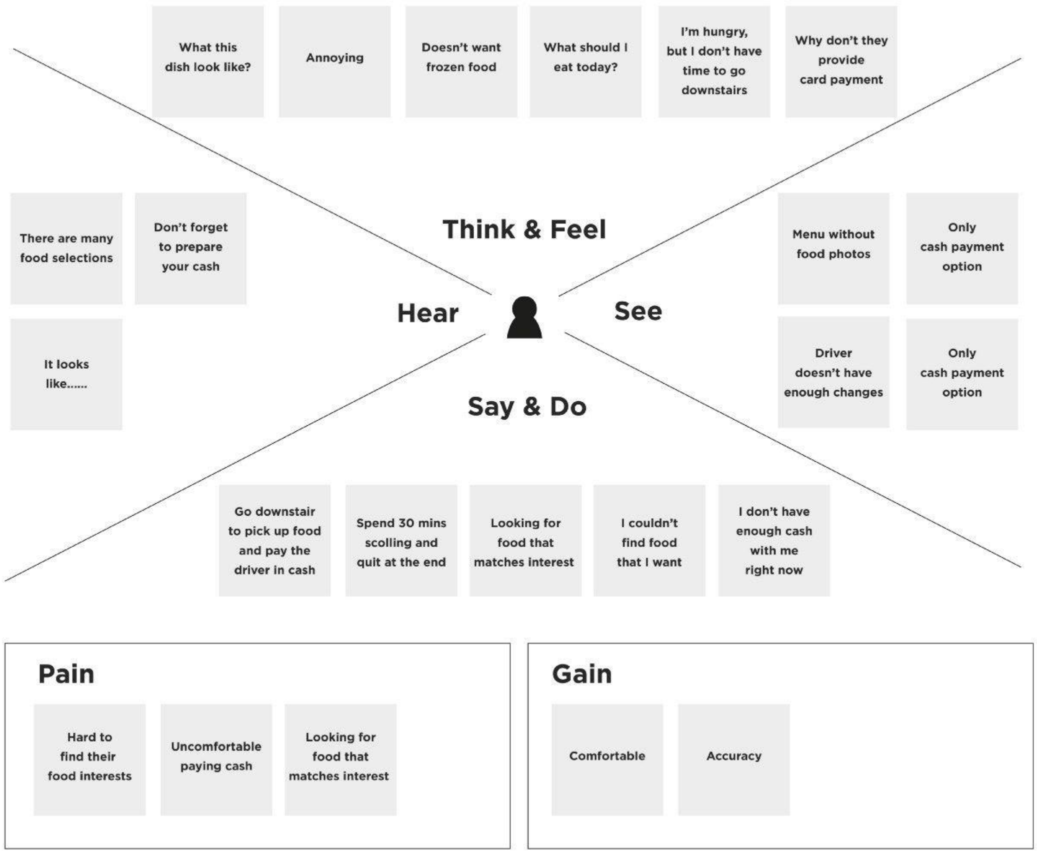
Empathy Map Canvas:
An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.
Example:



Reference: https://www.mural.co/templates/empathy-map-canvas

Example: Food Ordering & Delivery Application



FAIRPLANE

Guided city tours

Based on ten customer interviews and observations from the Fairplane Guided City Tours team

Claudia Larmon | Menaka Mahajan | Jerome Phillips | Alejandro Flores | Emma Sato

SCENARIO

Browsing, booking, attending, and rating a local city tour

Entice
How does someone initially become aware of this process?

Enter
What do people experience as they begin the process?

Engage
In the core moments in the process, what happens?

Exit
What do people typically experience as the process finishes?

Extend
What happens after the experience is over?

Steps
What does the person (or group) typically experience?

Entice
- **Booking other travel** — Most custom
- **Visit website or app** — A customer navigates to the city tours section of our website or app
- **Choose a city, dates, and number of people** — The customer types a city, dates, and the number of people who will attend the tour to see what tours are available
- **Browse available tours** — The customer sees available tours for their dates, city, and number of people
- **View detail on a single tour** — After seeing a tour that interests them, the customer clicks or taps to view more. They see information about what and where the tour will cover, plus its price, time of day, and tour guide.

Enter
- **Start purchase of a tour** — After deciding to go on this tour, they click the Purchase button
- **Complete payment information** — They fill out their contact and credit card information, then continue
- **Confirm payment & book tour** — They see a
- **Email confirmation** — An email immediately sends to confirm their tour and provide details about where and when to meet their guide
- **Email reminder** — One day before the tour begins, a reminder email is sent to all tour participants. The email emphasizes where and when to meet, and what to bring (if applicable).

Engage
- **Arrive at tour location** — Using their
- **Meet the guide & group** — Tour participa
- **Experience the tour** — The guide

Exit
- **Leave the guide & group** — The guide wraps up the tour and everyone heads their separate ways
- **Prompt for review** — One hour
- **Writing & submitting review** — The tour participant writes a review and gives the tour a star-rating out of 5.

Extend
- **Tour appears in the user profile** — The completed tour appears on the "past experiences" area of a customer's profile with a few details on where the group went
- **Personalized recommendations** — Participation in the tour informs our backend recommendation systems, which the customer may experience via better personalization
- **Personalized tour offers** — The customer receives an email 14 days after their tour with personalized recommendations for other tours
- **Personalized tour suggestions after new travel booking** — When a past tour participant books new travel with us, we show them personalized tour recommendations in their arrival city.

Interactions
What interactions do they have at each step along the way?

- **People:** Who do they see or talk to?
- **Places:** Where are they?
- **Things:** What digital touchpoints or physical objects would they use?

Entice
- Travel booking section of the website, iOS app, or Android app
- City tours section of the website, iOS app, or Android app
- City tours section of the website, iOS app, or Android app
- City tours section of the website, iOS app, or Android app
- City tours section of the website, iOS app, or Android app
- The tour guide

Enter
- City tours section of the website, iOS app, or Android app
- Payment overlay within the website, iOS app, or Android app
- Payment overlay within the website, iOS app, or Android app
- Customer's email (software like Outlook or website like Gmail)
- Customer's email (software like Outlook or website like Gmail)

Engage
- Tour locations
- The custome
- Direct interactions with the guide, and potentially other group members
- Direct interactions with the guide, and potentially other group members
- Some tours
- Most commo

Exit
- Direct interactions with the guide, and potentially other group members
- Often takes
- Depending on the tour participant and guide, tipping/cash may be involved
- Customer's email (software like Outlook or website like Gmail)
- "Leave a review"
- To some degree

Extend
- Completed
- If other users interact with this person, they will see these completed tours also
- Recommendations span across website, iOS app, or Android app
- Customer's email (software like Outlook or website like Gmail)
- Post-purchase screens website, iOS app, or Android app

Goals & motivations
At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")

Entice
- Help me get this flight or hotel booked
- Help me have more fun or learn new things on my trip
- Help me avoid
- Help me see what they have to offer
- Help me understand what this tour is all about

Enter
- Help me commit to going on this tour
- Help me get through this payment part without too much hassle
- Help me feel
- Help me feel
- Help me make

Engage
- Help me feel
- Help me feel good about my decision to go on this tour and to feel welcome
- Help me make the most of my trip to this new place

Exit
- Help me leave the tour with good feelings and no awkwardness
- Help me spread the word about a great tour or provide watch-outs and feedback for one that was not so good

Extend
- Help me see what I've done before
- Help me see what I could be doing next
- Help me see ways to enhance my new trip

Positive moments
What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?

Entice
- It's fun to look
- Tour photos, videos, and explanations are exciting to see
- It's reassuring to red reviews written by past travelers

Enter
- Excitement about the purchase ("Here we go!")
- Current payment flow is very bare-bones and simple
- We've heard from several people that the reminder emails were essential, especially if they booked way in advance

Engage
- Our guides tend to be so good that people are reassured when they meet their guide
- People love the tour itself, we have a 98% satisfaction rating

Exit
- People generally leave tours feeling refreshed and inspired

Extend
- People like looking back on their past trips
- We think people like these recommendations because they have an extremely high engagement rate

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Negative moments
What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?

Entice
- People sometimes forget to put in their dates or number of people, which leads them to discover tours they can't actually attend
- Several people expressed "information overload" as they browse
- People express a bit of fear of commitment at this step

Enter
- Trepidation about the purchase ("I hope this will be worth it!")

Engage
- People expressed awkwardness about finding their guide in a public place
- Sometimes people are matched up with tour participants that they don't really like

Exit
- People are unclear whether a tip is necessary, especially for non-Americans on an American tour
- People feel peer pressure to tip a guide when someone else on the tour tips, leaving them feeling weird and bad if they don't
- Customers report feeling review fatigue
- We have very low review rates (15% of people review experiences and tours)
- People describe leaving a review as an arduous process

Areas of opportunity
How might we make each step better? What ideas do we have? What have others suggested?

Entice
- If you don't follow this path immediately after your booking, could we send a follow-up?
- Could we automatically carry over the city from your booking? (e.g. via a cookie)
- Make it easier to compare and shop for experiences without having to click on them
- Provide a simpler summary to avoid information overload
- Show highlights or common phrases from reviews, or Uber style "great guide" badges?

Engage
- How might we make our guides easily identifiable (via a distinctive hat or shirt color, for example)?

Exit
- How might we make it clear that tipping is appreciated but not necessary?
- How might we equip people to tip after the tour? (e.g. via Venmo or equivalent app)
- Could we A/B test different language to see what changes response rates?
- How might we progressively disclose the full review so that each step feels more simple?

Extend
- How might we help people celebrate and remember things they've done in the past?
- How might we extend the personal connection to the guide long after the tour is over?

Project Design Phase-II
Data Flow Diagram & User Stories

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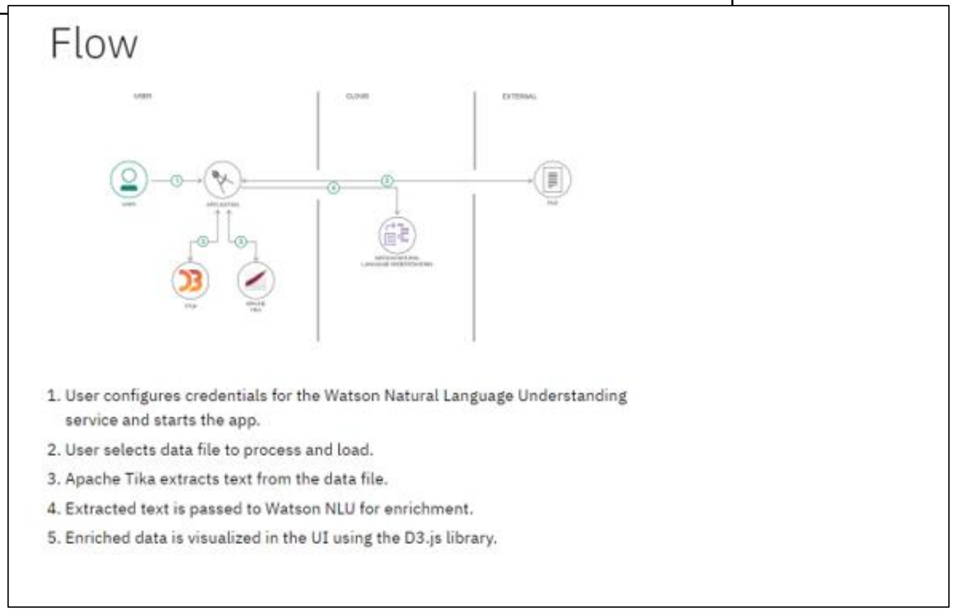
Maximum Marks	4 Marks

Data Flow Diagrams:
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: **(Simplified)**



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
Customer (Mobile user)	Registration	USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
Customer (Mobile user)	Registration	USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2
Customer (Mobile user)	Registration	USN-4	As a user, I can register for the application through Gmail	Dashboard is accessible via Gmail login	Medium	Sprint-1
Customer (Mobile user)	Login	USN-5	As a user, I can log into the application by entering email & password	User is redirected to the main dashboard	High	Sprint-1
Customer (Web user)	Dashboard	USN-6	As a student, I can view dietary habit insights on the dashboard.	Dashboard displays interactive charts successfully.	High	Sprint-2
Customer (Web user)	Data Analysis	USN-7	As a student, I can filter food data by category.	Data updates dynamically based on user filters.	High	Sprint-2
Administrator	Application Support	USN-8	As a support executive, I can resolve student queries regarding diet logs.	Support requests are resolved efficiently.	Medium	Sprint-3
Customer Care Executive	User Management	USN-9	As an administrator, I can manage student accounts and access.	User records are updated successfully.	High	Sprint-3
Administrator	System Monitoring	USN-10	As an administrator, I can monitor dashboard performance	Performance reports are available.	Medium	Sprint-3

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Food Choice Selection	Select food categories, view nutritional data
FR-4	Data Visualization	Display interactive Tableau charts , filter analysis
FR-5	Report Generation	Generate and download health analysis reports as PD
FR-6	Personalized feedback	View dietary habit analysis and health-based recommendations

Non-functional Requirements:
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Simple and user-friendly interface for students
NFR-2	**Security**	Student data and dietary logs must be securely stored
NFR-3	**Reliability**	Accurate and consistent food data analysis results
NFR-4	**Performance**	Dashboard visualization should load within 3 seconds
NFR-5	**Availability**	Application should be accessible 24*7 for students
NFR-6	**Scalability**	System should support multiple concurrent student users

Project Design Phase-II
Technology Stack (Architecture & Stack)

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Technical Architecture:
The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2
Example: Order processing during pandemics for offline mode
Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

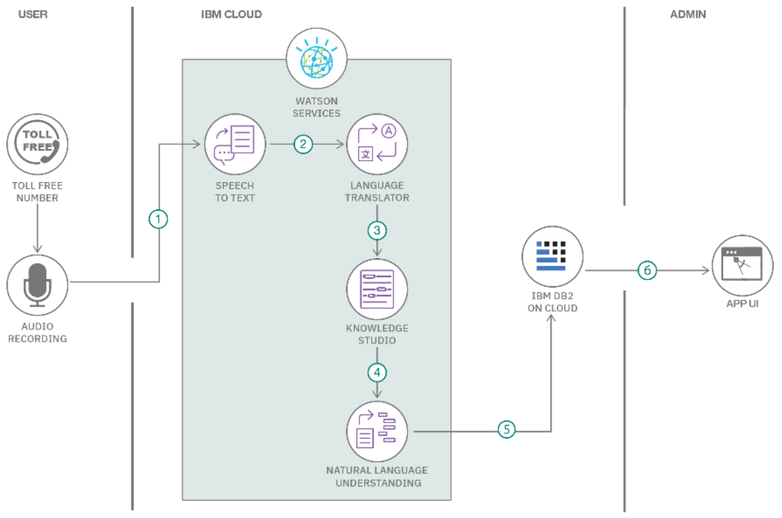

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc.	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Java / Python
3.	Application Logic-2	Logic for a process in the application	IBM Watson STT service
4.	Application Logic-3	Logic for a process in the application	IBM Watson Assistant
5.	Database	Data Type, Configurations etc.	MySQL, NoSQL, etc.
6.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.
7.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem

8.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
9.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
10.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN’s) etc.	Technology used

References:

https://c4model.com/
https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/
https://www.ibm.com/cloud/architecture
https://aws.amazon.com/architecture

Project Design Phase
Problem – Solution Fit Template

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Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) — CS

Who is your customer?
i.e. working parents of 0-5 y.o. kids

6. CUSTOMER CONSTRAINTS — CC

What constraints prevent your customers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

5. AVAILABLE SOLUTIONS — AS

Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.

9. PROBLEM ROOT CAUSE — RC

What is the real reason that this problem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

7. BEHAVIOUR — BE

What does your customer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS — TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

4. EMOTIONS: BEFORE / AFTER — EM

How do customers feel when they face a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

10. YOUR SOLUTION — SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

8. CHANNELS of BEHAVIOUR — CH

8.1 ONLINE
What kind of actions do customers take online? Extract online channels from #7

8.2 OFFLINE
What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

Extract online & offline CH of BE

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Project Design Phase
Proposed Solution Template

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Maximum Marks	2 Marks

Proposed Solution Template:
Project team shall fill the following information in the proposed solution template.

S.No.	**Parameter**	**Description**
1.	Problem Statement (Problem to be solved)	College students often struggle with making informed nutritional choices due to the lack of accessible and understandable dietary data.
2.	Idea / Solution description	Develop an interactive, web-based dashboard using Tableau and Flask to visualize student food consumption patterns and provide health-conscious insights.
3.	Novelty / Uniqueness	Combines real-world college food consumption datasets with interactive BI tools to offer personalized nutritional awareness at an institutional level.
4.	Social Impact / Customer Satisfaction	Promotes healthier eating habits among students, raising awareness about nutritional balance and long-term health outcomes.
5.	Business Model (Revenue Model)	Can be implemented as a health-tech service for college cafeterias or wellness programs through institutional subscriptions.
6.	Scalability of the Solution	The dashboard architecture allows for the integration of larger datasets from multiple campuses and the addition of predictive health analytics.

Project Design Phase
Solution Architecture

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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

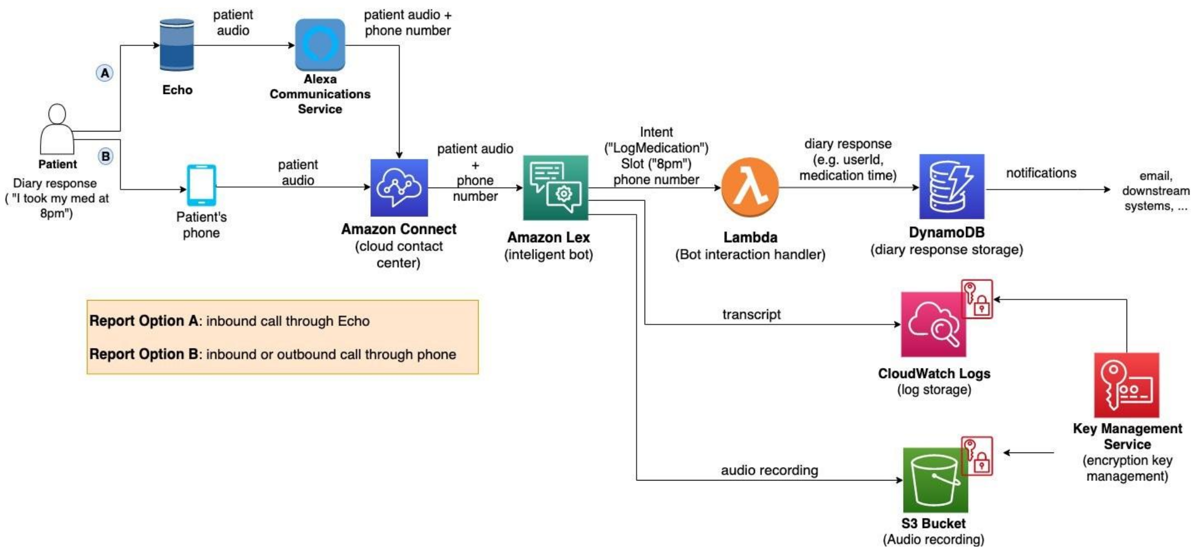

Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/

Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	Team
Sprint-1	Dashboard	USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	Team
Sprint-2	Application Support	USN-3	As a user, I can register for the application through Facebook	2	Low	Team
Sprint-1	System Monitoring	USN-4	As a user, I can register for the application through Gmail	2	Medium	Team
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	Team
	Dashboard					

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	3 Days	24 Jun 2026	26 Jun 2026	20	26 Jun 2026
Sprint-2	20	2 Days	27 Jun 2026	28 Jun 2026	20	28 Jun 2026
Sprint-3	20	5 Days	29 Jun 2026	30 Jun 2026	20	30 Jun 2026
Sprint-4	20	6 Days	01 Jul 2026	03 Jul 2026	20	03 Jul 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts

Functional & Performance Testing

Project Performance Test

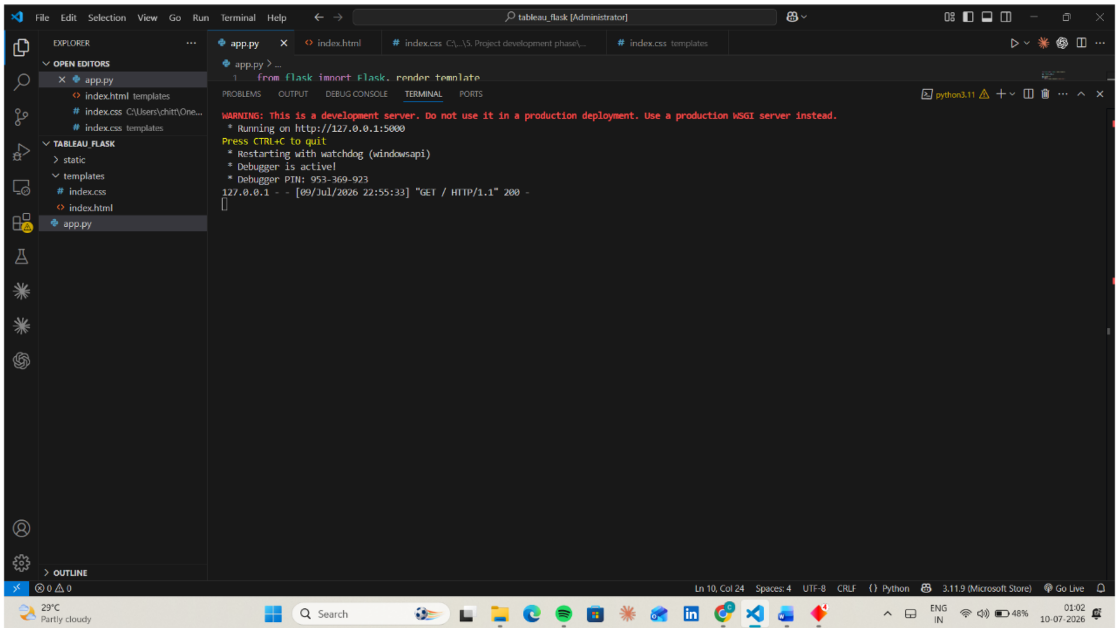
Date	**30 June 2026**
Team ID	**SWUID2026226489** **SWUID2026226490** **SWUID2026226275**
Project Name	**Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study**
Maximum Marks	**4 Marks**

Model Performance Testing:
Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Screenshot of Tableau Public connected to CSV source
2.	Data Preprocessing	Value: Cleaned and structured dietary dataset with 0 null values.
3.	Utilization of Filters	Screenshot of dynamic category/demographic filters applied.
4.	Calculation fields Used	Value: Applied 'Nutritional Index' and 'Calorie Count' calculated fields.
5.	Dashboard design	No of Visualizations / Graphs – 6 interactive charts
6	Story Design	No of Visualizations / Graphs -2 interactive story points

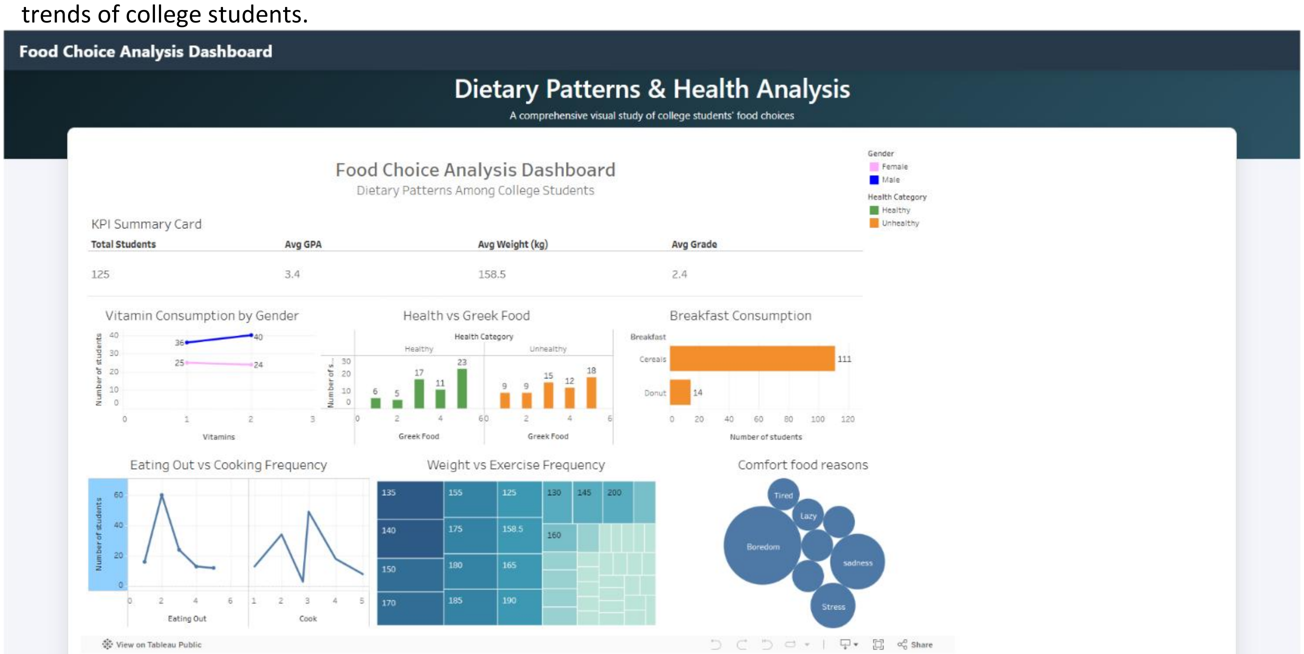
7.1 Output Screenshots

The **Comprehensive Analysis and Dietary Strategies with Tableau** project was successfully developed and tested using a Python-Flask web framework. The application integrates an interactive Tableau dashboard to visualize complex student dietary data. The system enables users to explore various parameters such as weight, exercise frequency, vitamin consumption, and comfort food preferences through dynamic, data-driven visual representations



Home Page

The home page of the **Food Choice Analysis Dashboard** provides an intuitive overview of the college food choices study. It introduces the primary purpose of the application, which is to analyze and visualize the dietary patterns and nutritional trends of college students.



8. ADVANTAGES & DISADVANTAGES

Advantages

1. **Data-Driven Insights**: Provides fast and accurate analysis of student dietary patterns and health correlations.Reduces manual effort in the loan approval process.
2. **Reduces Manual Effort**: Automates the visualization process, eliminating the need for manual data surveys and static reportEasy-to-use and user-friendly web interface.
3. **Improves Decision-Making**: Helps students and administrators make informed choices regarding health and nutritional well-beingSupports real-time loan prediction.
4. **Interactive Interface**: Offers an easy-to-use and intuitive web interface for exploring complex health metrics.
5. **Efficiency**: Saves significant time in data processing and trend identification for researchers and students.
6. **Real-Time Analytics**: Supports real-time exploration of student food choices through dynamic dashboard filters.

Disadvantages

Data Quality Dependency: The accuracy and reliability of the dashboard depend heavily on the quality and completeness of the student dietary dataset.

Maintenance: The dashboard may require updates or refinements when newer survey data or broader nutritional metrics become available.

User Input Errors: Inaccurate data entry during surveys can lead to skewed visualization results

Scope Limitation: The system provides analytical insights and cannot replace Of personalized medical or professional dietary consultation.

9. CONCLUSION

The **Comprehensive Analysis and Dietary Strategies with Tableau** project was successfully developed and tested using data visualization and web integration techniques. The application effectively analyzes and presents college students' dietary habits, providing quick and reliable visual insights. By automating data exploration, the project reduces manual analysis effort, improves institutional understanding of student health trends, and demonstrates the practical application of BI tools in the education and wellness sector..

10. FUTURE SCOPE

- Integrate real-time survey data collection forms directly into the dashboard.
- Develop advanced predictive analytics to forecast nutritional health trends among students.
- Add personalized user profiles for students to track their individual dietary progress.
- Deploy the application on cloud platforms for global accessibility.
- Create a mobile-optimized version for convenient access on the go.
- Generate automated monthly health recommendation reports for campus wellness programs.

11. APPENDIX

Source Code

The complete source code of the **A College Food Choices Case Study** is available in the GitHub repository.

GitHub Repository

https://github.com/chittlusai/A-College-Food-Choices-Case-Study

Project Demo Link:

https://drive.google.com/file/d/1o_ggb-P_eWY95L3oGC6l7Yn7KGLUHIgv/view?usp=sharing